

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

10 answers · Rationale-rich · Teach from doc

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Q1**A**

A. $x + y \geq 20$
 $3x + 4y \leq 80$

Choice A is correct. The total number of workbooks and sets of flash cards ordered is represented by $x + y$. Since the teacher must order at least 20 items, it must be true that $x + y \geq 20$. Each workbook costs \$3; therefore, $3x$ represents the cost, in dollars, of x workbooks. Each set of flashcards costs \$4; therefore, $4y$ represents the cost, in dollars, of y sets of flashcards. It follows that the total cost for x workbooks and y sets of flashcards is $3x + 4y$. Since the total cost of the order must not be over \$80, it must also be true that $3x + 4y \leq 80$. Of the choices given, these inequalities are shown only in choice A.

Choice B is incorrect. The second inequality says that the total cost must be greater, not less than or equal to \$80. Choice C incorrectly limits the cost by the minimum number of items and the number of items with the maximum cost. Choice D is incorrect. The first inequality incorrectly says that at most 20 items must be ordered, and the second inequality says that the total cost of the order must be at least, not at most, \$80.

Q2**4**

The correct answer is 4. It's given that line k is parallel to line j . It follows that the slope of line k is equal to the slope of line j . Given two points on a line in the xy -plane, x_1, y_1 and x_2, y_2 , the slope of the line can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. In the xy -plane shown, the points 0, 5 and 1, 9 are on line j . It follows that the slope of line j is $\frac{9 - 5}{1 - 0}$, or 4. Since the slope of line j is equal to the slope of line k , the slope of line k is also 4.

Q3**B**

B. 48

Choice B is correct. Multiplying both sides of the given equation by 4 yields $44x + 2 = 412$, or $16x + 8 = 48$. Therefore, the value of $16x + 8$ is 48.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**B**

B. 0, -4

Choice B is correct. The y-intercept of a graph is the point where the graph intersects the y-axis. The graph of function f shown intersects the y-axis at the point 0, -4. Therefore, the y-intercept of the graph is 0, -4.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**4**

The correct answer is 4. A solution to a system of equations must satisfy each equation in the system. It follows that if x, y is a solution to the system, the point x, y lies on the graph in the xy -plane of each equation in the system. According to the graph, the point x, y that lies on the graph of each equation in the system is 4, 1. Therefore, the solution to the system is 4, 1. It follows that the value of x is 4.

Q6**B****B.** 3.7

Choice B is correct. It's given that monarch butterflies can fly only with a body temperature of at least 55.0 degrees Fahrenheit °F. Let x represent the minimum increase needed in the monarch butterfly's body temperature to fly. If the monarch butterfly's body temperature is 51.3°F, the inequality $51.3 + x \geq 55.0$ represents this situation. Subtracting 51.3 from both sides of this inequality yields $x \geq 3.7$. Therefore, if the monarch butterfly's body temperature is 51.3°F, the minimum increase needed in its body temperature, in °F, so that it can fly is 3.7.

Choice A is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is 53.7°F, not 51.3°F.

Choice C is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is 50.0°F, not 51.3°F.

Choice D is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is 48.7°F, not 51.3°F.

Q7**B**

B. $10x + 5y = 700$

Choice B is correct. It's given that it takes the machine 10 minutes to make a large box. It's also given that x represents the possible number of large boxes the machine can make each day. Multiplying 10 by x gives $10x$, which represents the amount of time spent making large boxes. It's given that it takes the machine 5 minutes to make a small box. It's also given that y represents the possible number of small boxes the machine can make each day. Multiplying 5 by y gives $5y$, which represents the amount of time spent making small boxes. Combining the amount of time spent making x large boxes and y small boxes yields $10x + 5y$. It's given that the machine makes boxes for a total of 700 minutes each day. Therefore $10x + 5y = 700$ represents the possible number of large boxes, x , and small boxes, y , the machine can make each day.

Choice A is incorrect and may result from associating the time of 10 minutes with small, rather than large, boxes and the time of 5 minutes with large, rather than small, boxes.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q8**C**

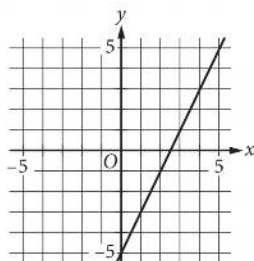
C. 18

Choice C is correct. It's given that $4x = 3$. Multiplying each side of this equation by 6 yields $24x = 18$. Therefore, the value of $24x$ is 18.

Choice A is incorrect. This is the value of $6x$, not $24x$.

Choice B is incorrect. This is the value of $8x$, not $24x$.

Choice D is incorrect. This is the value of $96x$, not $24x$.

Q9**D****D.**

Choice D is correct. In the xy -plane, the graph of the equation $y = mx + b$, where m and b are constants, is a line with slope m and y -intercept $(0, b)$. Therefore, the graph of $y = 2x - 5$ in the xy -plane is a line with slope 2 and a y -intercept $(0, -5)$. Having a slope of 2 means that for each increase in x by 1, the value of y increases by 2. Only the graph in choice D has a slope of 2 and crosses the y -axis at $(0, -5)$. Therefore, the graph shown in choice D must be the correct answer.

Choices A, B, and C are incorrect. The graph of $y = 2x - 5$ in the xy -plane is a line with slope 2 and a y -intercept at $(0, -5)$. The graph in choice A crosses the y -axis at the point $(0, 2.5)$, not $(0, -5)$, and it has a slope of $\frac{1}{2}$, not 2. The graph in choice B crosses the y -axis at $(0, -5)$;

however, the slope of this line is -2 , not 2. The graph in choice C has a slope of 2; however, the graph crosses the y -axis at $(0, 5)$, not $(0, -5)$.

Q10

C

C. 13

Choice C is correct. Adding the second equation of the given system to the first equation yields $5x + -4x + y = 15 + -2$, which is equivalent to $x + y = 13$. So the value of $x + y$ is 13.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $-(x + y)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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